

LANDFILL SITE SELECTION USING GIS IN ILORIN METROPOLIS, KWARA STATE, NIGERIA

The present study aim at determining land suitability for landfill site in Ilorin metropolis, Kwara State, for the disposal of solid wastes in the most efficient location. A significant amount of adverse environmental impact has resulted from an increasing increase in the generation of wastes and the lack of solutions to handle such materials, which have been observed in many regions, and particularly in Ilorin metropolis. This study proposes a methodology to optimize decision-making process to identify suitable sites for landfill using Geographical information system (GIS) and Multi criteria decision making (MCDM). Formulation of criteria for site selection and generation of criteria maps using GIS in order to identify, assess and perform spatial analysis in the study area.

The Study Area- Ilorin metropolis

Data to use :

In this study, the data to be use will include the location (point data) of the protected areas (mosques ,churches, and markets) and settlements, and also vector data (line) of the road network, water-course, other data used include digital elevation model (DEM), and Sentinel-2 resolution 10m(LULC), and Kwara state boundary shapefile (which include the study area), of which are accessible and gathered from different Open Datasets sources. Both the location data of the agricultural areas and the settlement were downloaded from the Gride3Nigeria (<https://grid3.gov.ng/>). Also, the road network data was also obtained from Gride3Nigeria (<https://grid3.gov.ng/>). Both the location Kwara state boundary shapefile and the Railway route data was obtained from DIVA-GIS([Free Spatial Data | DIVA-GIS](#)). While the surface slope was derived from the 30m resolution DEM downloaded from <http://earthexplorer.usgs.gov>, the DEM is a product of the Shuttle Radar Topography Mission (SRTM) which represents the bare ground topography, the slope through the DEM was extracted. Esri Sentinel-2 Land Cover of the study area were downloaded from ([http:// livingatlas.arcgis.com](http://livingatlas.arcgis.com)) through which the land use/land cover map was generated, table 3.1 presents data used and their sources. Table 3.1 presents the data used and their sources.

Euclidean distance analysis of landfill criteria

Proximity to road network

Proximity to settlements/residential area

Proximity to protected areas

Proximity to water course

Proximity to railway route

Land Use/ Land Cover

Surface Slope

METHODOLOGY : Analytical Hierarchical Process(AHP) Method